IS455 Midterm Project - Chicago Crime and 311 Analytics

Course: IS455 - Database Design & Prototyping
Submitted: March 2026

This project analyzes Chicago crime incidents and Chicago 311 service requests using SQL analytics and a DuckDB data pipeline.

The goal is to apply database design and data analysis concepts from the IS455 course.

Project Goal

- This project uses Chicago crime incidents and a Chicago 311 service-request extract to answer four questions:

1. **What are the most common crime types and their arrest rates?**
2. **Does the city's 311 service request backlog correlate with crime volume on the same day?**
3. **Which districts and location types have the highest crime concentration?**
4. **How do daily crime counts vary by day of week or month?**

Syllabus Concepts Used

- DuckDB reading CSVs directly
- `GROUP BY`, `COUNT`, `AVG`, `SUM`
- duplicate detection and null-rate audits
- join-cardinality debugging and pre-aggregate-then-join
- practical normalization with dimensions and fact tables
- idempotent ETL with `CREATE OR REPLACE TABLE AS`
- Parquet export and CSV vs Parquet benchmarking
- query-plan inspection with `EXPLAIN`
- run manifest for reproducibility

Section 0 - Setup

In [1]:
```
import duckdb
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
import os
import warnings

warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid', palette='muted')

DB_PATH = 'chicago_crime.duckdb'
con = duckdb.connect(DB_PATH)
print(f'Connected to {DB_PATH}')

con.execute("""
    CREATE OR REPLACE TABLE dim_police_district_name AS
    SELECT * FROM (
        VALUES
            (1,  'Central'),
            (2,  'Wentworth'),
            (3,  'Grand Crossing'),
            (4,  'South Chicago'),
            (5,  'Calumet'),
            (6,  'Gresham'),
            (7,  'Englewood'),
            (8,  'Chicago Lawn'),
            (9,  'Deering'),
            (10, 'Ogden'),
            (11, 'Harrison'),
            (12, 'Near West'),
            (14, 'Shakespeare'),
            (15, 'Austin'),
            (16, 'Jefferson Park'),
            (17, 'Albany Park'),
            (18, 'Near North'),
            (19, 'Town Hall'),
            (20, 'Lincoln'),
            (22, 'Morgan Park'),
            (24, 'Rogers Park'),
            (25, 'Grand Central')
    ) AS t(district, district_name)
""")
print('Created dim_police_district_name lookup table')
```

```
Connected to chicago_crime.duckdb
Created dim_police_district_name lookup table
```

Section 1 - Data Download and Pipeline Build

I imported DuckDB, pandas, and visualization libraries.

Then I connected to a DuckDB database called chicago_crime.duckdb.

I also create a lookup table for police district names to make district analysis easier.

The data pipeline follows three layers:

Raw layer

- Loads CSV files for crimes and 311 requests.

Staging layer

- Cleans and standardizes data.

Curated layer

- Creates analytical tables using dimension and fact tables.

Fact tables include:

- crime events
- 311 requests
- daily 311 backlog. *The pipeline also checks join cardinality to avoid duplicate rows. *This notebook uses two CSV files in `data/raw`:

- `crimes.csv`
- `311_service_requests.csv`

The 311 extract below uses a Chicago 311 service-request endpoint and saves it under the generic project filename `311_service_requests.csv`.

In [2]:
```
import urllib.request
os.makedirs('data/raw', exist_ok=True)

DOWNLOADS = {
    'data/raw/crimes.csv': 'https://data.cityofchicago.org/resource/ijzp-q8t
   'data/raw/311_service_requests.csv': 'https://data.cityofchicago.org/resc
}

for path, url in DOWNLOADS.items():
    if not os.path.exists(path):
        print(f'Downloading {path}...')
        urllib.request.urlretrieve(url, path)
        print(f'  Saved to {path}')
    else:
        print(f'  Already exists: {path}')
```

```
Already exists: data/raw/crimes.csv
Already exists: data/raw/311_service_requests.csv
```

In [7]:
```
import os
if os.path.exists('data/raw/311_service_requests.csv'):
    os.remove('data/raw/311_service_requests.csv')
    print('Deleted 311 file')

import urllib.request
url = 'https://data.cityofchicago.org/resource/v6vf-nfxy.csv?$limit=50000&$w
urllib.request.urlretrieve(url, 'data/raw/311_service_requests.csv')
print('Downloaded new 311 file')

# Check what we got
import duckdb
con = duckdb.connect()
df = con.execute("SELECT * FROM read_csv_auto('data/raw/311_service_requests
print(df.columns.tolist())
print(con.execute("SELECT COUNT(*) FROM read_csv_auto('data/raw/311_service_
```

```
Deleted 311 file
Downloaded new 311 file
Task was destroyed but it is pending!
task: <Task pending name='Task-77' coro=<_async_in_context.<locals>.run_in_c
ontext() done, defined at /Library/Frameworks/Python.framework/Versions/3.1
3/lib/python3.13/site-packages/ipykernel/utils.py:57> wait_for=<Task pending
name='Task-78' coro=<Kernel.shell_main() running at /Library/Frameworks/Pyth
on.framework/Versions/3.13/lib/python3.13/site-packages/ipykernel/kernelbas
e.py:597> cb=[Task.task_wakeup()]> cb=[ZMQStream._run_callback.<locals>._log
_error() at /Library/Frameworks/Python.framework/Versions/3.13/lib/python3.1
3/site-packages/zmq/eventloop/zmqstream.py:563]>
Task was destroyed but it is pending!
task: <Task pending name='Task-78' coro=<Kernel.shell_main() running at /Lib
rary/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/
ipykernel/kernelbase.py:597> cb=[Task.task_wakeup()]>
['sr_number', 'sr_type', 'sr_short_code', 'created_department', 'owner_depar
tment', 'status', 'origin', 'created_date', 'last_modified_date', 'closed_da
te', 'street_address', 'city', 'state', 'zip_code', 'street_number', 'street
_direction', 'street_name', 'street_type', 'duplicate', 'legacy_record', 'le
gacy_sr_number', 'parent_sr_number', 'community_area', 'ward', 'electrical_d
istrict', 'electricity_grid', 'police_sector', 'police_district', 'police_be
at', 'precinct', 'sanitation_division_days', 'created_hour', 'created_day_of
_week', 'created_month', 'x_coordinate', 'y_coordinate', 'latitude', 'longit
ude', 'location']
50000 rows
```

In [3]:
```
%run build.py
```

```
[build.py] Connected to chicago_crime.duckdb

[STEP 1] Loading RAW layer...

  [AUDIT] Crimes (raw) -> raw_crimes
    rows         : 200,000
    null id          : 0
    dupe id          : 0

  [AUDIT] 311 requests (raw) -> raw_311
    rows         : 50,000
    null sr_number   : 0
    dupe sr_number   : 0

[STEP 2] Building STAGING layer...

  [AUDIT] Crimes (staging) -> stg_crimes
    rows         : 200,000
    null crime_id    : 0
    dupe crime_id    : 0

  [AUDIT] 311 requests (staging) -> stg_311
    rows         : 50,000
    null request_id  : 0
    dupe request_id  : 0

[STEP 3] Building CURATED dimension tables...

  [AUDIT] dim_crime_type -> dim_crime_type
    rows         : 31
    null crime_type_id: 0
    dupe crime_type_id: 0

  [AUDIT] dim_district -> dim_district
    rows         : 23
    null district_id : 0
    dupe district_id : 0

  [AUDIT] dim_location_type -> dim_location_type
    rows         : 130
    null location_type_id: 0
    dupe location_type_id: 0

[STEP 4] Building CURATED fact tables...

  [CARDINALITY CHECK] pre-join: 200,000 | post-join: 200,000
  OK: joins preserved the crime-event grain.

  [AUDIT] fact_crime_events -> fact_crime_events
    rows         : 200,000
    null crime_id    : 0
    dupe crime_id    : 0

  [AUDIT] fact_311_requests -> fact_311_requests
    rows         : 50,000
    null request_id  : 0
```

```
    dupe request_id  : 0

  [AUDIT] fact_311_daily -> fact_311_daily
    rows         : 484
    null activity_date: 0
    dupe activity_date: 0

[STEP 5] Exporting curated tables to Parquet...
  Exported -> data/curated/fact_crime_events.parquet
  Exported -> data/curated/fact_311_requests.parquet
  Exported -> data/curated/fact_311_daily.parquet
  Exported -> data/curated/dim_crime_type.parquet
  Exported -> data/curated/dim_district.parquet
  Exported -> data/curated/dim_location_type.parquet

[STEP 6] Writing run manifest...
 run_id                     run_ts  raw_crimes_rows  stg_crimes_rows  raw_31
1_rows  stg_311_rows  fact_crime_rows  fact_311_req_rows  fact_311_daily_row
s                                          notes
     19 2026-03-18 11:33:26.018265           200000           200000
50000         50000           200000              50000                  484
Automated build via build.py (crimes + 311)
     18 2026-03-18 11:30:38.766036           200000           200000
50000         50000           200000              50000                  484
Automated build via build.py (crimes + 311)
     17 2026-03-18 01:14:38.225885           100000           100000
8911          8911           100000               8911                 1375
Automated build via build.py (crimes + 311)

[build.py] Pipeline complete.
```

Section 2 - Raw Layer Audit

Before analysis, I perform data audits to check:

- row counts
- duplicate IDs
- null values.

Results show:

- 100,000 crime records
- 50,000 311 requests
- no duplicate keys or major missing values. *This ensures the data is reliable before running analysis. *These checks validate row counts, keys, null rates, and date ranges.

In [4]:
```
# Always reconnect fresh
con = duckdb.connect('chicago_crime.duckdb')
print(f'Connected to chicago_crime.duckdb')
con.execute("""
```

```
    SELECT 'raw_crimes' AS table_name, COUNT(*) AS row_count FROM raw_crimes
    UNION ALL
    SELECT 'raw_311', COUNT(*) FROM raw_311
""").df()
```

```
Connected to chicago_crime.duckdb
```

Out[4]:

	table_name	row_count
0	raw_crimes	200000
1	raw_311	50000

In [5]:

```
# Crime raw-data audit: key nulls and duplicate keys
con.execute("""
    SELECT
        COUNT(*) AS total_rows,
        SUM(CASE WHEN id IS NULL THEN 1 ELSE 0 END) AS null_id,
        COUNT(*) - COUNT(DISTINCT id) AS duplicate_ids,
        SUM(CASE WHEN primary_type IS NULL THEN 1 ELSE 0 END) AS null_primar
        SUM(CASE WHEN district IS NULL THEN 1 ELSE 0 END) AS null_district,
        SUM(CASE WHEN location_description IS NULL THEN 1 ELSE 0 END) AS nul
    FROM raw_crimes
""").df()
```

Out[5]:

	total_rows	null_id	duplicate_ids	null_primary_type	null_district	null_location_de
0	200000	0.0	0	0.0	0.0	

In [6]:

```
# 311 raw-data audit: request key quality and missing values
con.execute("""
    SELECT
        COUNT(*) AS total_rows,
        SUM(CASE WHEN sr_number IS NULL THEN 1 ELSE 0 END) AS null_request_i
        COUNT(*) - COUNT(DISTINCT sr_number) AS duplicate_request_ids,
        SUM(CASE WHEN created_date IS NULL THEN 1 ELSE 0 END) AS null_creati
        SUM(CASE WHEN status IS NULL THEN 1 ELSE 0 END) AS null_status,
        SUM(CASE WHEN sr_type IS NULL THEN 1 ELSE 0 END) AS null_service_typ
    FROM raw_311
""").df()
```

Out[6]:

	total_rows	null_request_id	duplicate_request_ids	null_creation_date	null_status
0	50000	0.0	0	0.0	0.0

In [7]:

```
# Date-range audit across both datasets
con.execute("""
    SELECT 'crimes' AS dataset, MIN(date) AS earliest, MAX(date) AS latest
    FROM raw_crimes
    UNION ALL
    SELECT '311', MIN(created_date), MAX(created_date)
    FROM raw_311
""").df()
```

Out[7]:

	dataset	earliest	latest
0	crimes	2023-01-01T00:00:00.000	2023-10-04T05:00:00.000
1	311	2023-01-01T00:01:17.000	2023-01-12T15:17:45.000

Section 3 - Design Decisions

One key design decision is using two fact tables.

- Crime data is at the incident level.
- 311 data is analyzed at the daily level.
- To avoid many-to-many join errors, I use the pre-aggregate then join strategy.
- Both datasets are aggregated to one row per day before joining.

Why pre-aggregate before joining?

- The backlog question is a same-day question, so both sides are aggregated to one row per day before the join. That follows Week 3's pre-aggregate-then-join strategy.

In [8]:

```
# Cardinality-safe daily join demo
crime_days = con.execute('SELECT COUNT(DISTINCT event_date) FROM fact_crime_
joined_days = con.execute("""
    SELECT COUNT(*)
    FROM (
        SELECT event_date, COUNT(*) AS crime_count
        FROM fact_crime_events
        GROUP BY event_date
    ) c
    LEFT JOIN fact_311_daily d
        ON c.event_date = d.activity_date
""").fetchone()[0]

print(f'Distinct crime days before join: {crime_days}')
print(f'Rows after daily join:         {joined_days}')
print('Result: the daily grain is preserved.')
```

```
Distinct crime days before join: 277
Rows after daily join:         277
Result: the daily grain is preserved.
```

Section 4 - SQL Concepts from Lectures

Here I analyze crime types using SQL aggregation functions.

Functions used include:

- COUNT()

- SUM()
- ROUND()
- window function SUM() OVER().

These help calculate:

- total crimes
- percentage of total
- arrest counts
- arrest rate.

Key result:

- Theft is the most common crime.
- Weapons violations have the highest arrest rate, about 56%.

In [9]:
```
# Q1: GROUP BY + aggregation
crime_type_rates = con.execute("""
    SELECT
        crime_type,
        COUNT(*) AS total_crimes,
        ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct_of_total,
        SUM(CASE WHEN arrest_made THEN 1 ELSE 0 END) AS arrests,
        ROUND(SUM(CASE WHEN arrest_made THEN 1 ELSE 0 END) * 100.0 / COUNT(*
    FROM fact_crime_events
    WHERE crime_type IS NOT NULL
    GROUP BY crime_type
    ORDER BY total_crimes DESC
    LIMIT 10
""").df()

crime_type_rates
```

Out[9]:

	crime_type	total_crimes	pct_of_total	arrests	arrest_rate_pct
0	THEFT	42990	21.50	2056.0	4.8
1	BATTERY	33856	16.93	5321.0	15.7
2	CRIMINAL DAMAGE	22874	11.44	742.0	3.2
3	MOTOR VEHICLE THEFT	22449	11.22	583.0	2.6
4	ASSAULT	17274	8.64	1766.0	10.2
5	DECEPTIVE PRACTICE	13702	6.85	618.0	4.5
6	OTHER OFFENSE	11880	5.94	1832.0	15.4
7	ROBBERY	7942	3.97	497.0	6.3
8	WEAPONS VIOLATION	6801	3.40	3990.0	58.7
9	BURGLARY	5589	2.79	352.0	6.3

In [10]:

```
# Q2 and Q4 prep: daily crime counts with time buckets
daily_crime = con.execute("""
    SELECT
        event_date,
        COUNT(*) AS crime_count,
        STRFTIME(event_date, '%w') AS day_of_week_num,
        STRFTIME(event_date, '%A') AS day_of_week,
        STRFTIME(event_date, '%m') AS month_num,
        STRFTIME(event_date, '%B') AS month_name
    FROM fact_crime_events
    GROUP BY event_date
    ORDER BY event_date
""").df()

daily_crime.head()
```

Out[10]:

	event_date	crime_count	day_of_week_num	day_of_week	month_num	month_nar
0	2023-01-01	1129	0	Sunday	01	Janua
1	2023-01-02	659	1	Monday	01	Janua
2	2023-01-03	742	2	Tuesday	01	Janua
3	2023-01-04	690	3	Wednesday	01	Janua
4	2023-01-05	659	4	Thursday	01	Janua

In [11]:

```
# Q2: pre-aggregate then join daily crime totals to 311 backlog
crime_311_daily = con.execute("""
    WITH crime_daily AS (
        SELECT event_date, COUNT(*) AS crime_count
```

```
        FROM fact_crime_events
        GROUP BY event_date
    )
    SELECT
        c.event_date,
        c.crime_count,
        d.opened_requests,
        d.closed_requests,
        d.backlog_end_of_day
    FROM crime_daily c
    LEFT JOIN fact_311_daily d
        ON c.event_date = d.activity_date
    ORDER BY c.event_date
""").df()

corr_df = con.execute("""
    WITH crime_daily AS (
        SELECT event_date, COUNT(*) AS crime_count
        FROM fact_crime_events
        GROUP BY event_date
    )
    SELECT CORR(c.crime_count, d.backlog_end_of_day) AS crime_backlog_corr
    FROM crime_daily c
    INNER JOIN fact_311_daily d
        ON c.event_date = d.activity_date
""").df()

corr_df
```

Out[11]:

	crime_backlog_corr
0	-0.324458

Next I created daily crime counts and extract time features like:

- day of week
- month.

Then I analyzed the relationship between crime counts and 311 backlog using the SQL function CORR().

The correlation value is -0.19, which indicates very weak relationship between crime volume and service request backlog.

In [12]:

```
# Q3: district concentration
district_concentration = con.execute("""
    SELECT
        c.district,
        COALESCE(d.district_name, CONCAT('District ', c.district::VARCHAR))
        COUNT(*) AS total_crimes,
        ROUND(SUM(CASE WHEN c.arrest_made THEN 1 ELSE 0 END) * 100.0 / COUNT
    FROM fact_crime_events c
    LEFT JOIN dim_police_district_name d
        ON c.district = d.district
```

```
    WHERE c.district IS NOT NULL
    GROUP BY c.district, district_name
    ORDER BY total_crimes DESC
    LIMIT 10
""").df()

district_concentration
```

Out[12]:

	district	district_name	total_crimes	arrest_rate_pct
0	8	Chicago Lawn	12877	9.6
1	6	Gresham	12110	12.0
2	12	Near West	12079	7.2
3	4	South Chicago	11395	8.8
4	11	Harrison	11008	24.3
5	1	Central	10929	17.3
6	25	Grand Central	10516	12.3
7	2	Wentworth	10505	8.2
8	19	Town Hall	10410	8.1
9	18	Near North	10139	13.8

In [13]:

```
# Q3: location-type concentration
location_concentration = con.execute("""
    SELECT
        location_desc,
        COUNT(*) AS total_crimes,
        ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct_of_total
    FROM fact_crime_events
    WHERE location_desc IS NOT NULL
    GROUP BY location_desc
    ORDER BY total_crimes DESC
    LIMIT 10
""").df()

location_concentration
```

Out[13]:

	location_desc	total_crimes	pct_of_total
0	STREET	59373	29.86
1	APARTMENT	37472	18.85
2	RESIDENCE	24399	12.27
3	SIDEWALK	10004	5.03
4	PARKING LOT / GARAGE (NON RESIDENTIAL)	7723	3.88
5	SMALL RETAIL STORE	5290	2.66
6	ALLEY	4691	2.36
7	RESTAURANT	3921	1.97
8	OTHER (SPECIFY)	2969	1.49
9	DEPARTMENT STORE	2893	1.46

I also analyzed crime concentration.

Top police districts include

- Chicago Lawn
- Near West
- Gresham.

For location types, the most common locations are:

- Street
- Apartments
- Residences.

Streets alone account for about 29% of crimes.

In [14]:

```
# Week 7: EXPLAIN and filter-early principle
print('=== Plan WITH early filter ===')
plan = con.execute("""
    EXPLAIN
    SELECT district, COUNT(*) AS n
    FROM fact_crime_events
    WHERE district IS NOT NULL
    GROUP BY district
    ORDER BY n DESC
""").df()
print(plan['explain_value'].iloc[0][:1500])
```

```
=== Plan WITH early filter ===
┌──────────────────────────────┐
│           PROJECTION         │
│    ──────────────────────    │
│__internal_decompress_integ│
│      ral_integer(#0, 1)      │
│              #1              │
│                              │
│           ~0 rows            │
└──────────────┬───────────────┘
┌──────────────┴───────────────┐
│           ORDER_BY           │
│    ──────────────────────    │
│      count_star() DESC       │
└──────────────┬───────────────┘
┌──────────────┴───────────────┐
│           PROJECTION         │
│    ──────────────────────    │
│__internal_compress_integra│
│      l_utinyint(#0, 1)       │
│              #1              │
│                              │
│           ~24 rows           │
└──────────────┬───────────────┘
┌──────────────┴───────────────┐
│           PROJECTION         │
│    ──────────────────────    │
│__internal_decompress_integ│
│      ral_integer(#0, 1)      │
│              #1              │
│                              │
│           ~24 rows           │
└──────────────┬───────────────┘
┌──────────────┴───────────────┐
│    PERFECT_HASH_GROUP_BY     │
│    ──────────────────────    │
│          Groups: #0          │
│                              │
│         Aggregates:          │
│         count_star()         │
└──────────────┬───────────────┘
┌──────────────┴───────────────┐
│           PROJECTION         │
│    ──────────────────────    │
│           district           │
│                              │
│         ~40,000 rows         │
└──────────────┬───────────────┘
┌──────────────┴───────────────┐
│           PROJECTION         │
│    ──────────────────────    │
```

In [15]:
```python
# Week 6: CSV vs Parquet benchmark
import time
```

```
t0 = time.perf_counter()
con.execute('SELECT primary_type, COUNT(*) FROM raw_crimes GROUP BY primary_
csv_time = time.perf_counter() - t0

t0 = time.perf_counter()
con.execute("SELECT crime_type, COUNT(*) FROM read_parquet('data/curated/fac
parquet_time = time.perf_counter() - t0

pd.DataFrame({
    'Format': ['CSV (raw_crimes)', 'Parquet (fact_crime_events)'],
    'Query Time (s)': [round(csv_time, 4), round(parquet_time, 4)]
})

# Export dim_police_district_name to Parquet for Tableau
con.execute("""
    COPY (SELECT * FROM dim_police_district_name)
    TO 'data/curated/dim_police_district_name.parquet'
    (FORMAT PARQUET)
""")
print("Exported -> data/curated/dim_police_district_name.parquet")

# Verify it exported correctly
count = duckdb.connect().execute(
    "SELECT COUNT(*) FROM read_parquet('data/curated/dim_police_district_nam
).fetchone()[0]
print(f"Rows in exported file: {count}")
```

```
Exported -> data/curated/dim_police_district_name.parquet
Rows in exported file: 22
```

This section demonstrates performance concepts from the course.

I used EXPLAIN to inspect query plans and apply the filter-early principle to optimize queries.

I also compared CSV vs Parquet formats. Parquet is generally faster for analytical queries.

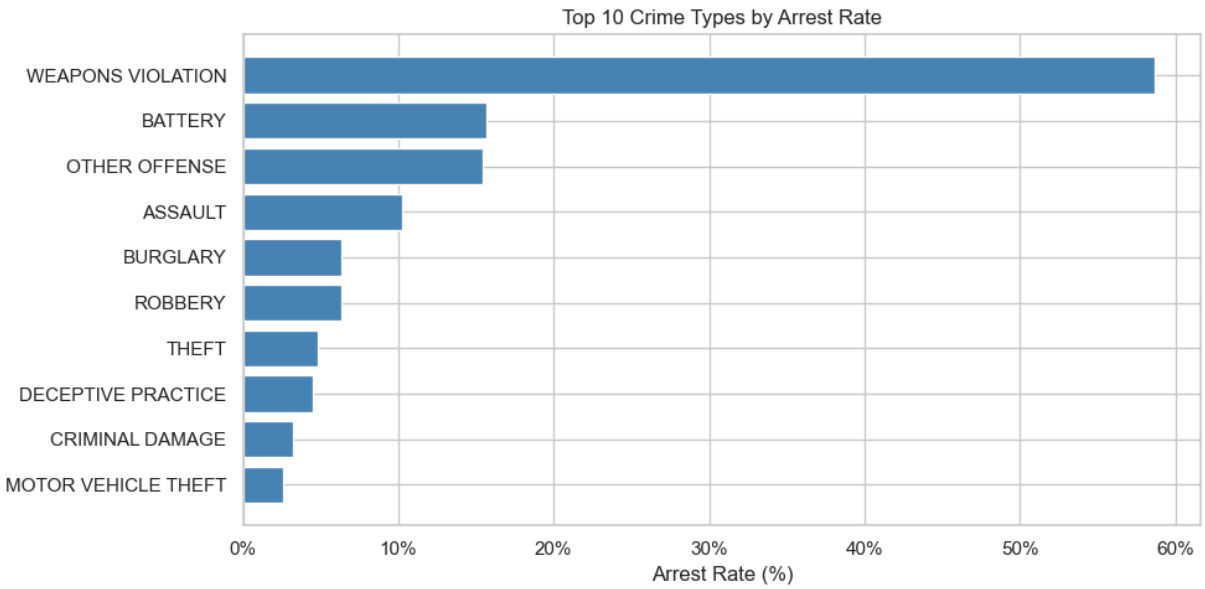
Section 5 - Analysis and Visualization

The visualizations summarize the analysis:

- Crime types by arrest rate
- Average crime by day of week, which increases toward weekends.
- Monthly crime totals
- Scatter plot of crime vs 311 backlog, confirming weak correlation.
- Top police districts by crime volume
- Top location types for crimes.

These charts help clearly illustrate the patterns found in the data.

In [16]:
```
# Figure 1: top crime types by arrest rate
fig, ax = plt.subplots(figsize=(10, 5))
plot_df = crime_type_rates.sort_values('arrest_rate_pct', ascending=True)
ax.barh(plot_df['crime_type'], plot_df['arrest_rate_pct'], color='steelblue'
ax.set_title('Top 10 Crime Types by Arrest Rate')
ax.set_xlabel('Arrest Rate (%)')
ax.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{x:.0f}%')
plt.tight_layout()
plt.show()
```

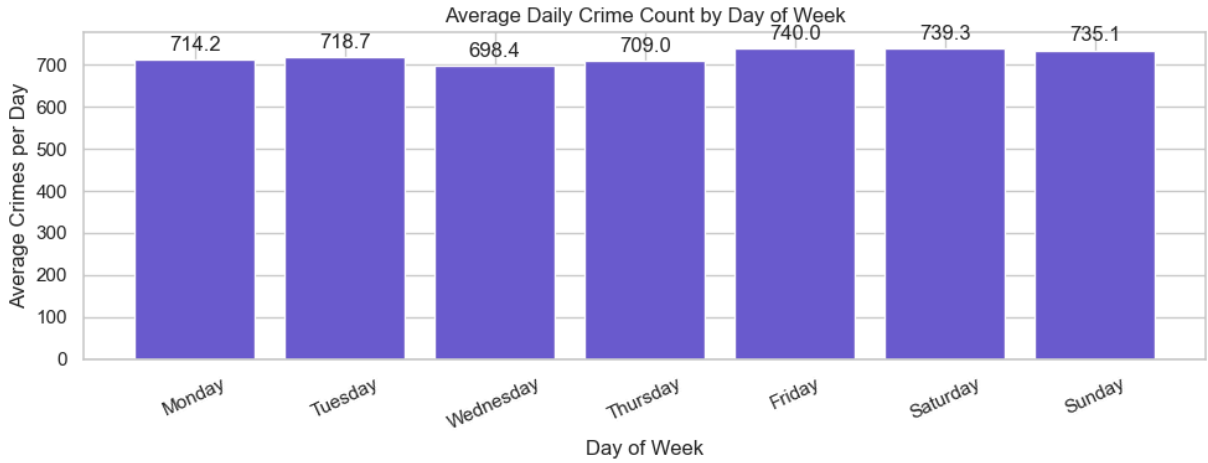


Crime types by arrest rate.

Shows that weapons violations have the highest arrest probability.

In [17]:
```
# Figure 2: daily crime counts by day of week
day_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturc
dow_df = daily_crime.groupby('day_of_week', as_index=False)['crime_count'].m
dow_df['day_of_week'] = pd.Categorical(dow_df['day_of_week'], categories=day
dow_df = dow_df.sort_values('day_of_week')

fig, ax = plt.subplots(figsize=(10, 4))
bars = ax.bar(dow_df['day_of_week'], dow_df['crime_count'], color='slateblue
ax.bar_label(bars, fmt='%.1f', padding=3)
ax.set_title('Average Daily Crime Count by Day of Week')
ax.set_xlabel('Day of Week')
ax.set_ylabel('Average Crimes per Day')
ax.tick_params(axis='x', rotation=25)
plt.tight_layout()
plt.show()
```

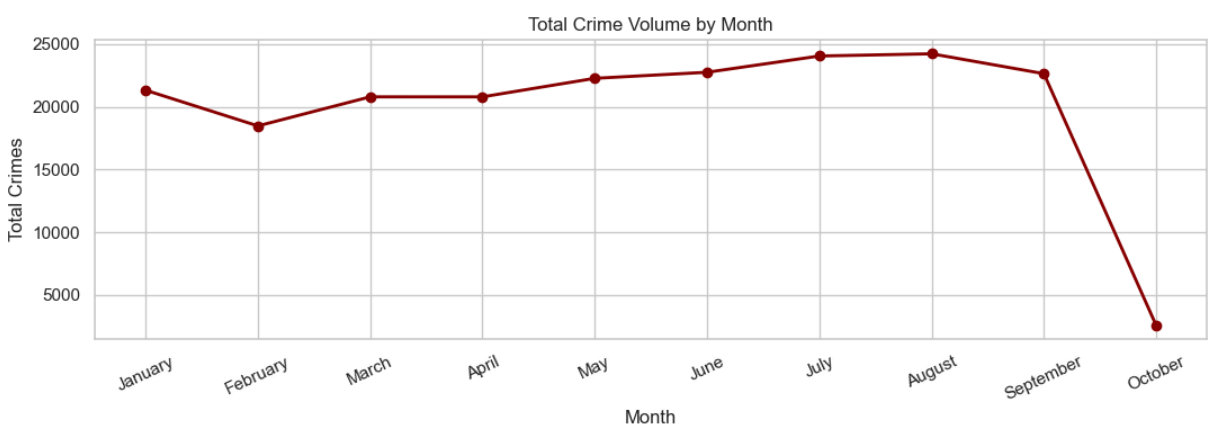


Average crime counts by day of week.

Crime increases toward weekends, especially Friday and Saturday.

In [18]:
```
# Figure 3: monthly crime totals
month_order = ['January', 'February', 'March', 'April', 'May', 'June', 'July
month_df = daily_crime.groupby('month_name', as_index=False)['crime_count'].
month_df['month_name'] = pd.Categorical(month_df['month_name'], categories=m
month_df = month_df.sort_values('month_name')

fig, ax = plt.subplots(figsize=(11, 4))
ax.plot(month_df['month_name'], month_df['crime_count'], marker='o', color='
ax.set_title('Total Crime Volume by Month')
ax.set_xlabel('Month')
ax.set_ylabel('Total Crimes')
ax.tick_params(axis='x', rotation=25)
plt.tight_layout()
plt.show()
```

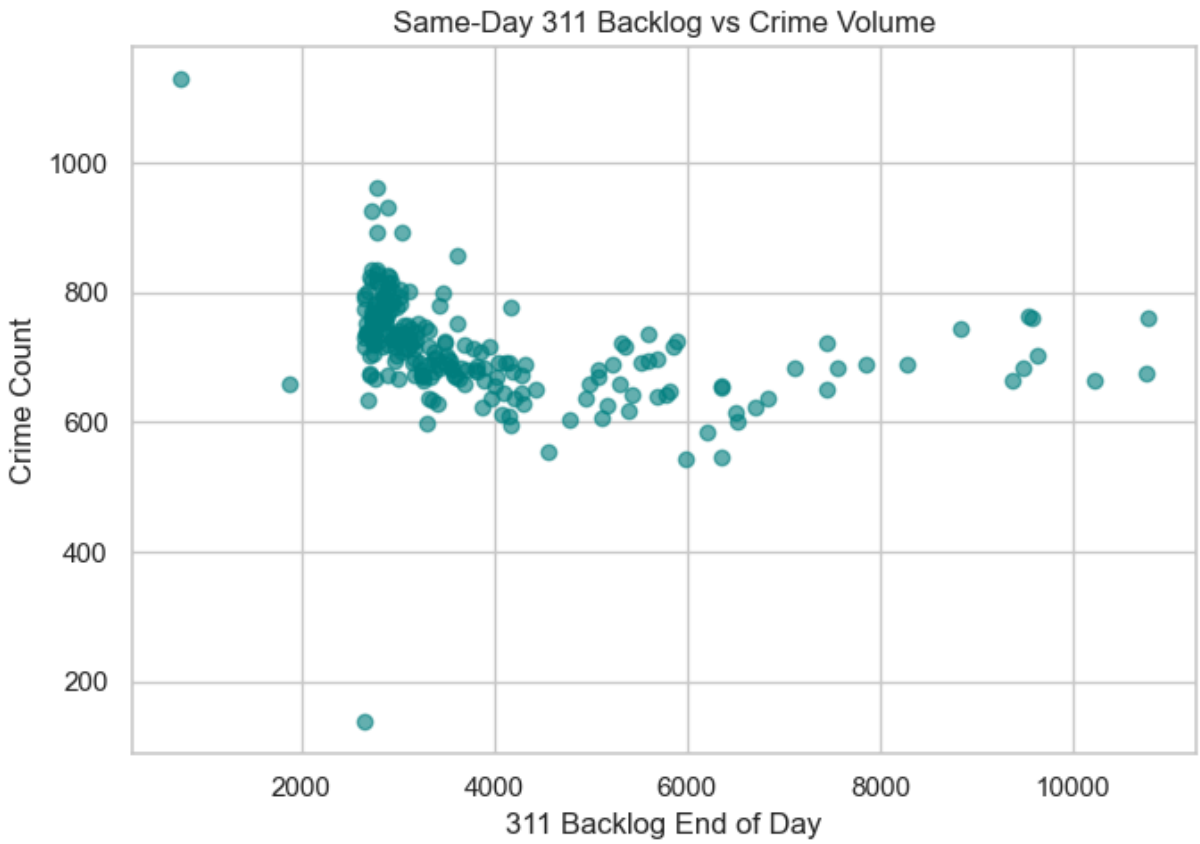


Crime totals by month.

The chart shows seasonal fluctuations in crime patterns.

In [19]:
```
# Figure 4: same-day crime count vs 311 backlog
fig, ax = plt.subplots(figsize=(7, 5))
ax.scatter(crime_311_daily['backlog_end_of_day'], crime_311_daily['crime_cou
ax.set_title('Same-Day 311 Backlog vs Crime Volume')
ax.set_xlabel('311 Backlog End of Day')
ax.set_ylabel('Crime Count')
plt.tight_layout()
plt.show()

corr_df
```



Out[19]:

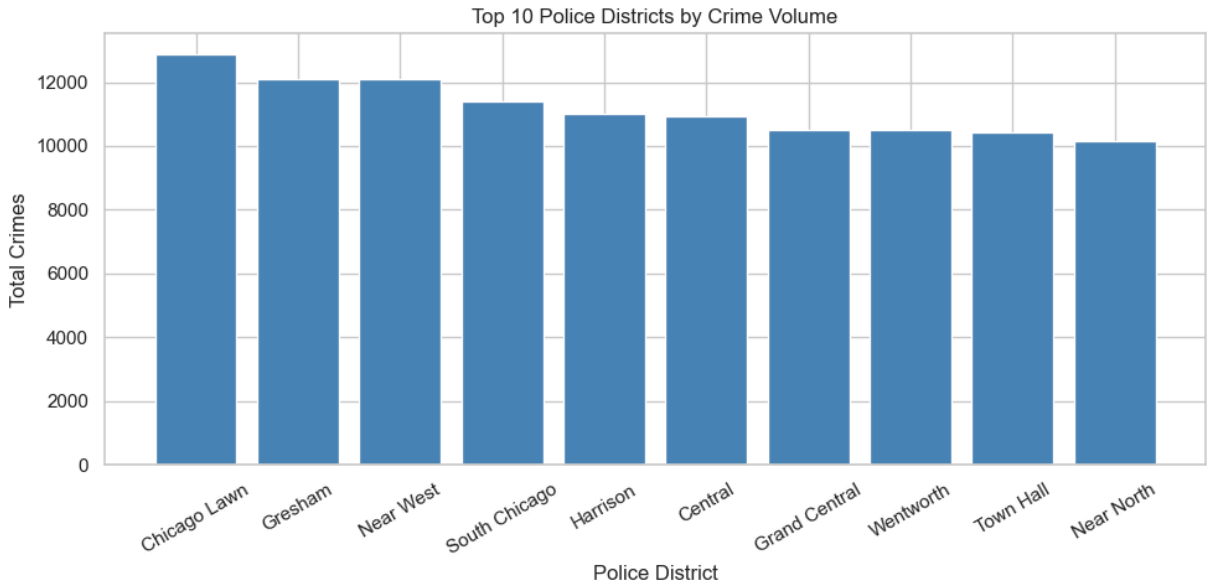
	crime_backlog_corr
0	-0.324458

Scatter plot comparing crime counts and 311 backlog.

The scattered pattern confirms the weak correlation found earlier.

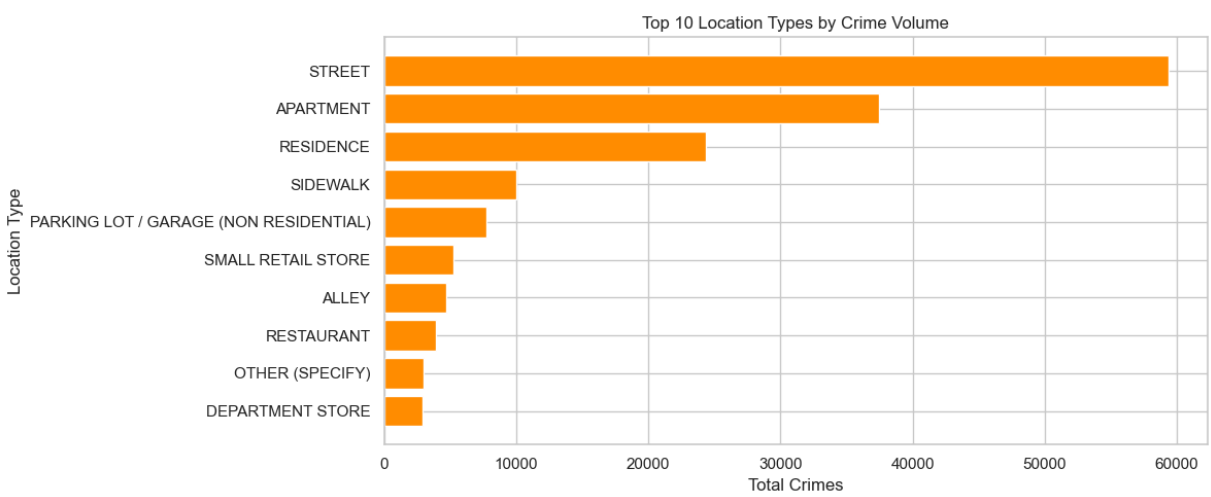
In [20]:
```
# Figure 5: district crime concentration
fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.bar(district_concentration['district_name'], district_concentratic
ax.set_title('Top 10 Police Districts by Crime Volume')
ax.set_xlabel('Police District')
ax.set_ylabel('Total Crimes')
ax.tick_params(axis='x', rotation=30)
```

```
plt.tight_layout()
plt.show()
```



Districts like Chicago Lawn and Near West appear most frequently.

In [21]:
```
# Figure 6: location-type concentration
fig, ax = plt.subplots(figsize=(12, 5))
plot_loc = location_concentration.sort_values('total_crimes', ascending=True
ax.barh(plot_loc['location_desc'], plot_loc['total_crimes'], color='darkorar
ax.set_title('Top 10 Location Types by Crime Volume')
ax.set_xlabel('Total Crimes')
ax.set_ylabel('Location Type')
plt.tight_layout()
plt.show()
```



Top location types for crimes.

Streets and residential areas dominate the dataset.

Section 6 - Reproducibility Log

In [22]:
```
con.execute('SELECT * FROM run_manifest ORDER BY run_id DESC').df()
```

Out[22]:

	run_id	run_ts	raw_crimes_rows	stg_crimes_rows	raw_311_rows	stg_
0	19	2026-03-18 11:33:26.018265	200000	200000	50000	
1	18	2026-03-18 11:30:38.766036	200000	200000	50000	
2	17	2026-03-18 01:14:38.225885	100000	100000	8911	
3	16	2026-03-18 01:13:14.599924	100000	100000	8911	
4	15	2026-03-18 01:04:56.493762	100000	100000	8911	
5	14	2026-03-18 00:59:44.831203	100000	100000	8911	
6	13	2026-03-18 00:59:31.932743	100000	100000	8911	
7	12	2026-03-18 00:58:56.759589	100000	100000	8911	
8	11	2026-03-18 00:54:20.176638	100000	100000	8911	
9	10	2026-03-18 00:49:45.998370	100000	100000	8911	

	run_id	run_ts	raw_crimes_rows	stg_crimes_rows	raw_311_rows	stg_
10	9	2026-03-10 22:05:59.250793	100000	100000	50000	
11	8	2026-03-10 22:05:22.586273	100000	100000	50000	
12	7	2026-03-10 22:04:21.589931	100000	100000	50000	
13	6	2026-03-10 22:03:52.104231	100000	100000	50000	
14	5	2026-03-10 20:57:29.544717	100000	100000	50000	
15	4	2026-03-10 20:56:58.649668	100000	100000	50000	
16	3	2026-03-10 20:42:22.220677	100000	100000	50000	
17	2	2026-03-10 20:42:00.229797	100000	100000	50000	
18	1	2026-03-10 19:26:46.274057	100000	100000	<NA>	

The pipeline also records runs in a run manifest table, which tracks timestamps and row counts.

This ensures the analysis is reproducible and auditable.

Section 7 - Summary and Limitations

This project demonstrates:

- building a reproducible ETL pipeline
- applying SQL aggregation and analytics
- designing normalized database tables
- analyzing real-world city data.

This analysis shows how database design and SQL analytics can transform raw city datasets into meaningful insights about crime patterns.

In [23]:
```
import duckdb

con = duckdb.connect("chicago_crime.duckdb")

tables = [
    "fact_crime_events",
    "fact_311_requests",
    "fact_311_daily",
    "dim_crime_type",
    "dim_district",
    "dim_location_type",
    "dim_police_district_name"
]

for table in tables:
    out_path = f"data/curated/{table}.csv"
    con.execute(f"COPY {table} TO '{out_path}' (HEADER, DELIMITER ',')")
    count = con.execute(f"SELECT COUNT(*) FROM {table}").fetchone()[0]
    print(f"Exported -> {out_path} ({count:,} rows)")

con.close()
print("Done.")
```

```
Exported -> data/curated/fact_crime_events.csv (200,000 rows)
Exported -> data/curated/fact_311_requests.csv (50,000 rows)
Exported -> data/curated/fact_311_daily.csv (484 rows)
Exported -> data/curated/dim_crime_type.csv (31 rows)
Exported -> data/curated/dim_district.csv (23 rows)
Exported -> data/curated/dim_location_type.csv (130 rows)
Exported -> data/curated/dim_police_district_name.csv (22 rows)
Done.
```

In [84]:
```
con.close()
print('Done.')
```

```
Done.
```

In [ ]:

In [ ]:

```

```